MarketPulse: Stock Return Distribution & Risk Analytics System

ENGINEERING CAPSTONE PROJECT – 1 (23IE4053R / 23IE4053A)

Project Review-1

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Under the Guidance of

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Outline of Project Work

- Focus
- Introduction
- Literature Survey
- Challenges
- Research Objectives
- Conclusion & Future Work
- References
- Publications Details

Problem Statement

Most people checking a stock look at its price chart and its average return. Neither tells you how risky the stock really is. A share can look steady for months and then lose a big chunk in a few bad days, and the average hides all of that. Real return data also has fatter tails than the normal curve suggests, so extreme losses happen more often than textbook models expect.

Risk measures like volatility, skewness, kurtosis, drawdown and Value at Risk exist, but beginners rarely see them explained in plain words. Many tools either throw raw numbers at the user or try to predict prices, which is unreliable. Very few check whether their own risk estimates actually held up on past data.

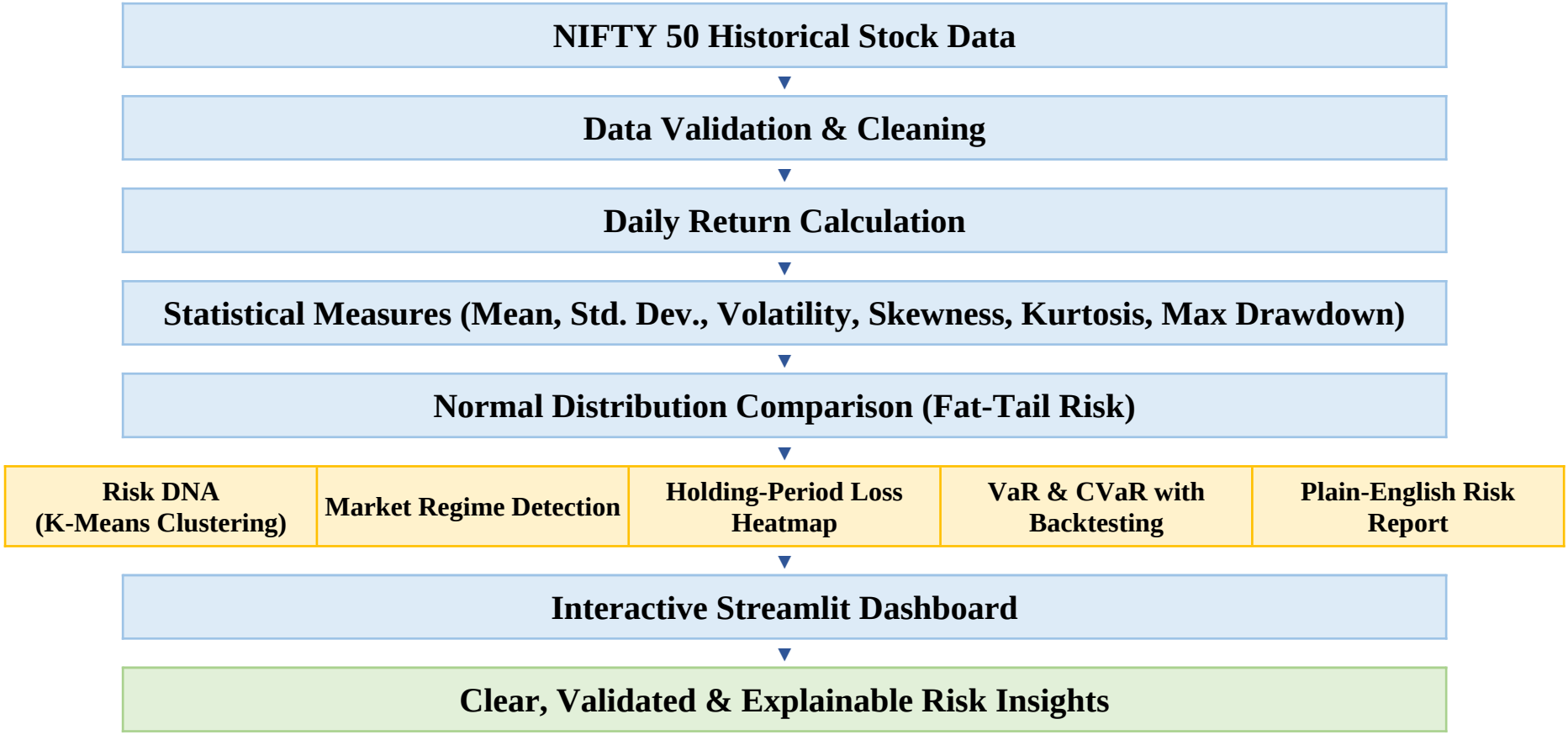
MarketPulse is our answer to this. Using historical NIFTY 50 data, it measures return risk statistically, compares returns with the normal distribution, groups stocks by risk personality, detects market regimes, estimates VaR and CVaR with backtesting, and writes the findings in plain English on a Streamlit dashboard. It is built to explain past risk, not to forecast prices.

Objectives

- Collect and clean historical NIFTY 50 stock data for analysis.
- Compute daily returns, mean, standard deviation, volatility, skewness, kurtosis and maximum drawdown.
- Compare return distributions with the normal distribution to spot fat-tail risk.
- Group stocks into risk personalities (Risk DNA) using K-Means clustering.
- Classify market periods as calm, normal, turbulent or crisis and compare returns across them.
- Build a holding-period loss probability heatmap for different investment durations.
- Estimate Value at Risk and Conditional VaR, and validate them with backtesting.
- Generate an automated plain-English risk report from the numerical results.
- Deliver everything through an interactive Streamlit dashboard.

PROJECT DESIGN

MarketPulse Workflow



Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Benoit Mandelbrot	"The Variation of Certain Speculative Prices" Journal of Business, 1963	• Examined how speculative price changes are distributed, using cotton prices.	• Large changes occur far more often than a normal distribution suggests.	• His proposed model is hard to use in everyday analysis; later work refined it.
Rama Cont	"Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues" Quantitative Finance, 2001	• Collected common statistical features of return data across many markets.	• Returns show heavy tails, volatility clustering and weak autocorrelation.	• Describes the patterns but gives no ready-to-use tool for students or investors.
James D. Hamilton	"A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle" Econometrica, 1989	• Used a regime-switching model to let a series move between distinct states.	• Behaviour can differ sharply from one regime to another, so one set of statistics may not fit all periods.	• Applied to economic output; needs adapting for daily stock returns.
R. Tyrrell Rockafellar, Stanislav Uryasev	"Optimization of Conditional Value-at-Risk" Journal of Risk, 2000	• Studied CVaR, the average loss beyond the VaR level, as a risk measure.	• CVaR captures tail losses that VaR alone ignores.	• Focused on portfolio optimization; relies on good tail estimates.
Paul H. Kupiec	"Techniques for Verifying the Accuracy of Risk Measurement Models" Journal of Derivatives, 1995	• Proposed a test based on how often actual losses exceed the VaR estimate.	• A VaR model should be validated by backtesting, not simply trusted.	• The test has low power when the sample is small.

Dataset – NIFTY 50 Historical Data

Dataset Details

- Source: Publicly available historical price data of NIFTY 50 stocks (for example, Yahoo Finance)
- Domain: Indian stock market, financial risk analytics
- Records: Daily trading data for the selected stocks over a multi-year period
- Raw fields: Date, Open, High, Low, Close, Adjusted Close and Volume
- Derived fields: Daily return, rolling volatility, drawdown, VaR and market regime label
- Target: None for price. Stocks are clustered into risk groups, and no prices are predicted

Example Attributes

- Closing price
- Daily return
- Maximum drawdown
- Rolling volatility
- Skewness and kurtosis
- Market regime
- VaR and CVaR
- Holding period
- Cluster (Risk DNA)

Problem Gap Identification

Most beginner tools stop at price charts and average returns, and say little about how returns are distributed.

Real returns have fat tails, but few tools show this against the normal distribution.

Stocks are usually treated alike, with no grouping by how they actually behave in risk terms.

One set of statistics is applied to all periods, even though calm and crisis markets behave very differently.

VaR numbers are often shown without any backtesting to check if they can be trusted.

Numbers are rarely turned into plain-language explanations for students and beginner investors.

Gap addressed by MarketPulse

MarketPulse combines distribution analysis, Risk DNA clustering, regime detection, loss-probability heatmaps, validated VaR/CVaR and plain-English reports in one educational dashboard, without pretending to predict the market.

Innovation, Creativity, and Novelty

1. Risk DNA

K-Means clustering on distribution-based features groups stocks into risk personalities.

2. Market Regime Detection

Labels periods as calm, normal, turbulent or crisis and compares return distributions across them.

3. Holding-Period Loss Probability Heatmap

Shows the historical chance of loss for different investment durations.

4. VaR and CVaR with Backtesting

Estimates potential loss and its tail average, then tests the estimates against past data.

5. Automated Plain-English Risk Report

Converts the numerical results into insights anyone can read.

6. Interactive Streamlit Dashboard

Brings all of the above into one browser interface with no coding needed.

Feasibility and Project Plan

Technical Feasibility

Uses well-known Python libraries: Pandas, NumPy, SciPy, scikit-learn and Plotly.

NIFTY 50 historical data is freely available and easy to handle.

Streamlit lets us build the dashboard with very little code.

Operational Feasibility

Runs in a normal web browser, so anyone can use it.

Users select a stock and get charts, risk figures and a written report right away.

No special hardware or paid software is needed.

Economic Feasibility

Relies on open-source tools and free data sources.

Daily data for 50 stocks is small, so clustering and backtesting run on an ordinary laptop.

A low-cost prototype that fits comfortably within an academic project.

Project Plan

Phase	Topics	Work
Phase 1	Setup	Requirements, NIFTY 50 data source and Python environment
Phase 2	Data	Collecting, cleaning and validating the stock price data
Phase 3	Analysis	Returns, volatility, skewness, kurtosis, maximum drawdown and normal-distribution comparison
Phase 4	Novel Features	Risk DNA clustering, regime detection, loss heatmap, VaR/CVaR with backtesting
Phase 5	Report & App	Plain-English risk report and Streamlit dashboard with Plotly charts
Phase 6	Integration	Joining all modules and testing with different NIFTY 50 stocks
Phase 7	Finalization	Quality checks, documentation and presentation